Tomáš Gonda

Researcher in Applied Category Theory and Quantum Information Theory

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Education

Perimeter Institute & University of Waterloo, Physics, Quantum Information Theory — Waterloo, ON, Canada, Sept 2017 – Nov 2021
- Thesis: *Resource Theories as Quantale Modules*
- Supervisor: Dr. Robert W. Spekkens

Perimeter Institute & University of Waterloo, Theoretical Physics — Waterloo, ON, Canada, Aug 2016 – June 2017
- Thesis: *Resource Theoretic Approach to Causal Inference*
- Supervisor: Dr. Robert W. Spekkens

University of Edinburgh, Mathematics & Physics — Edinburgh, Scotland, Oct 2013 – May 2016
- First class honours, 92% average grade
- Thesis: *Symplectic Geometry*
- Supervisor: Dr. Johan Martens

Experience

University of Innsbruck, Senior postdoctoral researcher — Innsbruck, Austria, Mar 2025 – present, 1 year 1 month

Independent research in applied category theory and quantum information theory.
- Algebra group, led by Tim Netzer
- Funded by my FWF ESPRIT Grant Ergodic Theory in Categorical Probability

University of Innsbruck, Postdoctoral researcher — Innsbruck, Austria, Jan 2022 – Feb 2025, 3 years 2 months

Conducting research in applied category theory and quantum information theory.
- Mathematical quantum physics group, led by Gemma De Les Coves

Applied Category Theory Adjoint School, Supervisor / teaching assistant — June 2024

Co-leading a summer school research project with Paolo Perrone.
- Topic: *Modelling Uncertainty with Markov Categories*

University of Waterloo & University of Innsbruck, Supervisor / project lead — 2022 – present, 4 years

Co-supervising Master's thesis projects of:
- Yìlè Yīng (MSc, 2022, co-supervised by RW Spekkens); *Asymmetry Trade-off Relations: A New Kind of Uncertainty Relations*
- Stefan Kremminger (MSc, 2024); *Quantum Conditional Independence*

University of Waterloo, Teaching assistant — Waterloo, ON, Canada, 2017 – 2021, 4 years
- Conducting tutorials for graduate level courses in Quantum Theory and Statistical Physics.
- Supervising multiple Master's level winter school projects in Quantum Foundations.

ETH Zürich, Visiting researcher — Zürich, Switzerland, Summer 2019

Collaborating with researchers in the group of Renato Renner on topics in information theory.

InstaEDU, Online tutor — Remote, 2013 – 2016, 3 years

Online tutoring of undergraduate students in science and engineering.

University of Edinburgh, Career development internship — Edinburgh, Scotland
A summer project on modelling turbulence in Matlab. — Summer 2015

- Topic: *Transition to turbulence in plane Couette flow*

Personal tutor — Bratislava, Slovakia
In-person tutoring of high school students. — 2012 – 2013
1 year

Volunteering

Positive Futures Festival, Organizer — Innsbruck, Austria
Co-organizing an annual two week long music festival with ~40 concerts and ~1000 attendees. — 2024 – present

Touch&Play Cologne Festival, Organizer — Germany
Co-organizing and facilitating an annual week long retreat with ~100 attendees. — 2025 – present

Awards

ESPRIT Grant Ergodic Theory in Categorical Probability — Mar 2025
The ESPRIT Grant supports young independent researchers in Austria.
Austrian Science Fund
www.fwf.ac.at/forschungsradar/10.55776/ESP3451824

Napier Medal and Gangadhar Balwant Gadgil Prize — 2016
Awarded to the best student in the Year 4 of the Mathematics program at the University of Edinburgh.
University of Edinburgh

Nichol Foundation Scholarship — 2015
Awarded to the best overall mark in the Year 3 of Physics degrees at the University of Edinburgh.
University of Edinburgh

Kelland Memorial Prize — 2014
Awarded for distinguished performance for direct entrants into the Second year of the program.
University of Edinburgh